

moment about mean describes the symmetry or skewness (i.e., lack of symmetry) of the distribution of a random variable.

We state a few important theorems on moments without going into details as they have been covered in the unit on probability.

- 1) $\sigma^2 = \mu'_2 - \mu^2$
- 2) If the random variable X has the variance σ^2 , then $\text{Var}(aX + b) = a^2\sigma^2$
- 3) **Chebyshev's theorem:** To determine how σ or σ^2 is indicative of the spread or the dispersion of the distribution of the random variable, Chebyshev's theorem is very useful. Here we will only state the theorem:

If μ and σ^2 are the mean and variance of a random variable, say X , then for any constant k

$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2; \sigma \neq 0$$

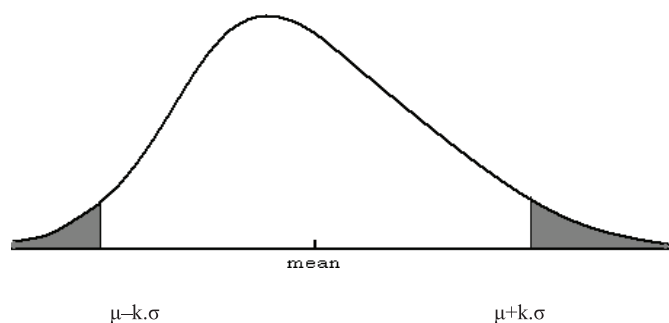


Fig 26.5: Chebyshev's Theorem

26.6.2 Moment Generating Functions

Although moments of most distributions can be determined directly by evaluating the necessary integrals or sums, there is an alternative procedure, which some times provides considerable simplifications. This technique is known as the technique of moment generating function.

The moment generating function of a random variable X , where it exists is given by,

$$M_X(t) = E(e^{tx}) = \sum_{i=1}^n e^{tx} f(x); \text{ when } X \text{ is discrete and}$$

$$M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx; \text{ when } X \text{ is continuous}$$

The independent variable is t , and we are interested in values of t in the neighbourhood of 0. To explain why, we refer to this function as a “moment generating function”. Let us substitute for e^{tx} with its Maclaurin's series expansion as follows:

$$e^{tx} = 1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^r x^r}{r!} + \dots$$

Thus, for discrete case we get,

$$\begin{aligned}
 M_x(t) &= E(e^{tx}) = \sum_{i=1}^n e^{tx} f(x) = \sum_{i=1}^n f(x) \sum_{i=1}^n e^{tx} f \\
 &= (1 + tx + t^2 x^2 / 2! + t^3 x^3 / 3! + \dots + t^r x^r / r! + \dots) \\
 &= \sum_{i=1}^n f(x) + t \sum_{i=1}^n x f(x) + t^2 / 2! \sum_{i=1}^n x^2 f(x) + t^3 / 3! \sum_{i=1}^n x^3 f(x) + \dots + t^r / r! \sum_{i=1}^n x^r f(x) + \dots \\
 &= 1 + t \cdot \mu + t^2 / 2! \cdot \mu^2 + \dots + t^r / r! \cdot \mu^r + \dots
 \end{aligned}$$

Thus, we can see that in the Maclaurin's series of the moment generating function of X, the coefficient of $t^r / r!$ is μ^r , which is nothing but the r^{th} order moment about the origin. In the continuous case, the argument is the same.

To get the r^{th} order moment about the origin, we differentiate $M_x(t)$ r times with respect to t and put $t=0$ in the expression obtained. Symbolically,

$$\mu_r' = d^r M_x(t) / dt^r |_{t=0}$$

An example will provide more clarity.

Example: Find the moment generating function of the random variable whose probability density is given by

$$F(x) = e^{-x} \text{ if } x > 0$$

$$0 \text{ otherwise}$$

and use that to find the expression for μ_r .

By definition,

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{+\infty} e^{tx} f(x) dx = \int_0^{\infty} e^{tx} e^{-x} dx = \int_0^{\infty} e^{-x(1-t)} dx = 1 / (1-t); \text{ for } t < 1$$

When $|t| < 1$, the Maclaurin's series for this moment generating function is

$$\begin{aligned}
 M_x(t) &= 1 + t + t^2 + t^3 + t^4 + \dots + t^r + \dots = 1 + 1! \cdot t / 1! + 2! \cdot t^2 / 2! + 3! \cdot t^3 / 3! + \dots + r! \cdot t^r / r! + \dots
 \end{aligned}$$

Hence, $\mu_r' = d^r M_x(t) / dt^r |_{t=0} = r!$ for $r = 0, 1, 2, \dots$

If 'a' and 'b' are constants, then

- 1) $M_{x+a}(t) = E(e^{t(x+a)}) = e^{at} \cdot M_x(t)$
- 2) $M_{x.b}(t) = E(e^{t.b.x}) = M_x(bt)$
- 3) $M(x+a)/b(t) = E(e^{t(x+a)/b}) = e^{(a/b)t} \cdot M_x(t/b)$

Among the above three results, the third one is of special importance. When $a = -\mu$ and $b = \sigma$, $M_{(x-\mu)/\sigma}(t) = E(e^{t(x-\mu)/\sigma}) = e^{(-\mu/\sigma)t} \cdot M_x(t/\sigma)$

Check Your Progress 2

- 26) Given X has the probability distribution $f(x) = 1/8 \cdot {}^3C_x$ for $x=0,1,2$, and 3, find the moment generating function of this random variable and use it to determine μ'_1 and μ'_2 .

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26.7 LET US SUM UP

In this unit, we learnt about the various concepts related to probability distribution. We understood how random variables are always generated with a particular pattern of probability attached to them. Based on that, we discussed the probability distributions and their properties. We defined the random variable and made a broad distinction of the probability distributions based on whether the random variable is continuous or discrete. Then we discussed how the moments of a probability distribution describes the distribution completely; how the technique of moment generating function could be used to obtain the moments of any probability distribution.

26.8 KEY WORDS

Continuous Random Variable	: If a random variable can take any value within its range, then it is called a continuous random variable.
Discrete Random Variable	: If a random variable takes only countable number of values and there are no possible values of the variable located between two juxtaposed values of the variable, then it is called a discrete random variable.
Probability Distribution	: Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities.
Probability Mass Function	: Often the probability of the discrete random variable X assuming the value x is represented by $f(x)$, $f(x) = P(X = x)$ = probability that X assumes value x . The function $f(x)$ is known as the probability mass function (p.m.f.) given that it satisfies following two conditions:

$$1) f(x) \geq 0$$

$$2) \sum_{i=1}^n f(x_i) = 1 \sum_{i=1}^n f(x)$$

Probability Density Function : A continuous random variable having a probability distribution $f(x)$ which is a continuous on-negative function, gives the probability that the random variable will lie in a specified interval when integrated over the interval, i.e.,

$$P(c \leq x \leq d) = \int_c^d f(x) dx$$

The function $f(x)$ is called the probability density function (p.d.f.) provided it satisfies following two conditions

$$1) f(x) \geq 0$$

2) If the range of the continuous random variable is

$$(a,b) \text{ integration } \int_a^b f(x) dx = 1$$

26.9 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A p.m.f. must satisfy the following two properties

$$i) f(x) \geq 0$$

$$ii) \sum_{i=1}^n f(x_i) = 1 \sum_{i=1}^n f(x_i) = 1$$

Using the second property, we get $10k^2 + 9k - 1 = 0$. It gives two values of k , viz., -1 and $1/10$. Clearly, k cannot take the value -1 (as $f(X = 1) = k$ and $f(x)$ is always non-negative). Given $k = 1/10$ rest is merely algebra.

2) i) Cannot, (since, $f(x=1) < 0$ and probability cannot be negative)

$$ii) \text{ Yes, can be a p.m.f since } (f(x) \geq 0 \text{ and } \sum_0^4 f(x) = 1$$

$$iii) \text{ Cannot since } (f(x)) \geq 0 \text{ but } \sum_0^4 f(x) \neq 1$$

Check Your Progress 2

$$1) M_X(t) = E(e^{tx}) = \sum_{i=0}^3 e^{tx} \frac{1}{8} C_X = 1/8 (1 + 3e^t + 3e^{2t} + e^{3t}) = 1/8 (1 + e^t)$$

$$\mu_1 = M_X(0) = 3/2$$

$$\mu_2 = M_X(0) = 3$$

26.10 EXERCISES

Q1. A fair coin is tossed twice. Let X be defined as the no. of heads shown. Find the range of X and its probability mass function P_X .

Ans. The sample space $S = \{HH, TT, HT, TH\}$

The no. of heads can be 0,1,or 2.

So, $R(x) = \{0,1,2\}$

The probability mass function $P_X(x) = P(X = x)$ for $x = 0,1,2$

$P_X(0) = P(X = 0) = \frac{1}{4}$ [i.e. probability of getting no head]

$P_X(1) = P(X = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

$P_X(2) = P(X = 2) = \frac{1}{4}$

Q2. From the below given p.m.f, find the value of r .

Y	0	1	2	3
P(Y=y)	0	2r	3r	6r ²

Ans. We know, $\sum P(y_i) = 1$.

Thus,

$$0 + 2r + 3r + 6r^2 = 1$$

$$5r + 6r^2 - 1 = 0$$

$$6r^2 + 6r - r - 1 = 0$$

$$6r(r + 1) - 1(r + 1) = 0$$

$$(6r - 1)(r + 1) = 0$$

So, $6r - 1 = 0$ and $r + 1 = 0$

$r = -1$ is not possible as the value of probability lies between 0 and 1.

Thus, $r = 1/6$

Therefore, the value of r from the above stated PMF is $1/6$.

Q3 What are the two conditions for probability density function?

Ans. The probability density function will only be valid if it meets the following conditions:

- 1) $f(x)$ should be non-negative for all values of the random variable.
- 2) The area underneath $f(x)$ should be equal to 1.

Q4. What is the difference between the probability mass function and the probability density function?

Ans. The main difference between the probability mass function (PMF) and the probability density function (PDF) is that PDF is used to describe the continuous probability distribution whereas PMF is used to describe the discrete probability distribution.

UNIT 27 PROBABILITY DISTRIBUTIONS-II

Structure

- 27.0 Objectives
- 27.1 Introduction
- 27.2 Joint Distribution
- 27.3 Marginal Distribution
- 27.4 Three Important Probability Distributions
 - 27.4.1 Binomial Distribution
 - 27.4.2 Poisson Distribution
 - 27.4.3 Normal Distribution
- 27.5 Let Us Sum Up
- 27.6 Key Words
- 27.7 Answers/Hints to Check Your Progress Exercises
- 27.8 Exercises

27.0 OBJECTIVES

After going through this unit, you should be able to

- discuss the concept of joint and marginal distribution;
- explain the three most common probability distributions (binomial, poisson and normal); and
- solve questions related to these three distributions.

27.1 INTRODUCTION

The previous unit has elaborated on the concept of random variables and the basics of probability distribution. This understanding would serve as the basis of the topics that we are going to cover in this unit. Uptil now, we have introduced the probability models involving only a single random variable. This unit will explain how probability modelling changes when we are interested in more than one random variable at the same time. Therefore, we would discuss about joint distribution, the associated marginal distribution and further the case of how to represent these when variables are independent. Additionally, three very commonly used probability distributions would be discussed, namely: binomial distribution, poisson distribution and normal distribution.

27.2 JOINT DISTRIBUTION

After having discussed about the random variables and the basics of probability distribution, we are now proceeding to the topic of joint distribution.

Very commonly, there might arise situations where we are dealing with more than one random variable. This concept of joint distribution allows us to understand the relationship between multiple events happening at the same time, e.g., from a deck of 52 cards, the joint probability of picking up a card that is both red and 6

Let us denote two events A and B and these be described by random variables X at value x and Y at value y. So, $P(A \cap B) = P(X = x \cap Y = y) = P(X = x, Y = y)$ and this expression is referred to as the joint probability of $X=x$ and $Y=y$.

More formal way to describe the same concept is as follows:

If X and Y are discrete random variables, then the function given by $f(x,y) = P(X=x, Y=y)$ for each pair of values (x,y) within the range of X is called the joint probability distribution of X and Y.

Thus a bivariate function can serve as the **joint probability distribution** of a pair of discrete random variables X and Y if and only if the values of $f(x, y)$ satisfies the condition:

- i) $f(x,y) \geq 0$ for each pair of values (x,y) within its domain
- ii) $\sum_x \sum_y f(x,y)$ where the double summation extends over all possible points (x,y) within its domain.

Example: Determine the value of k for which the function given by $f(x,y) = kxy$; for $x = 1, 2, 3$; $y = 1, 2, 3$ can serve as a joint distribution.

Ans. Substituting the values of x and y we get:
 $f(x, y = 1) = k$; $f(1,2) = 2k$; $f(1,3) = 3k$;
 $f(2,1) = 2k$; $f(2,2) = 4k$; $f(2,3) = 6k$; $f(3,1) = 3k$; and $f(3,2) = 6k$; $f(3,3) = 9k$

We now have to check the conditions for the joint probability distribution: $k > 0$ and

$$\sum_x \sum_y f(x,y) = k + 2k + 3k + 2k + 4k + 6k + 3k + 6k + 9k = 1$$

So that $36k = 1$ and $k = \frac{1}{36}$.

If we have more than two discrete random variables i.e., $X_1, X_2, \dots, X_n$, then the joint p.m.f of the variables is the function,

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

If the variables are continuous, the joint p.d.f of $X_1, \dots, X_n$ is the function $f(x_1, x_2, \dots, x_n)$ such that for any n intervals $[a_1, b_1], \dots, [a_n, b_n]$,

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \dots dx_1$$

Example: Assume we have a corpus of 100 words. Table 27.1 depicts the words and the associated frequencies and probabilities of each of those words.

Table 27.1: Frequencies and Probabilities of 100 words

W	c(w)	P(w)	X	y
Shall	5	0.05	5	1
Up	8	0.08	2	1
Next	5	0.05	4	1
As	5	0.05	2	1
Fore	3	0.03	4	2
To	10	0.1	2	1
We	12	0.12	2	1
Some	8	0.08	4	2
Will	30	0.3	4	1
BBC	14	0.14	3	0

Here random variable X denotes the length of the word and Y denotes the number of vowels in the word.

Now, let us calculate $f(x,y) = P(X=x, Y=y)$ taking different values of the random variable defined here (refer Table 27.2).

$$f(x = 4, y = 2) = P(\text{Fore}) + P(\text{Some}) = 0.03 + 0.08 = 0.11$$

$$f(x = 3, y = 0) = P(\text{BBC}) = 0.14$$

Table 27.2: Calculation of $f(x, y)$

		x			
		2	3	4	5
y	0	0	0.14	0	0
	1	0.35	0.3	0.05	0.05
	2	0	0	0.11	0

Moving forward, it becomes imperative to also mention about the special case of the theorem of joint distribution of independent random variables. According to this theorem, if X and Y are independent, then $f(x,y) = f_X(x) \cdot f_Y(y)$ for all values of x and y. This can also be stated in an alternative manner: We say that the random variables $X_1, X_2, \dots, X_n$ are independent if for every subset $X_{i_1}, X_{i_2}, \dots, X_{i_k}$ of the variables, the joint p.m.f or p.d.f of the subset is equal to the product of the marginal p.m.fs or p.d.fs.

If X and Y are discrete random variables the function given by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{x \leq s} \sum_{y \leq t} f(s, t); \text{ for } -\infty < X < \infty; -\infty < Y < \infty$$

where, $f(s, t)$ is the value of the joint probability distribution of X and Y at (s, t) is called the joint distribution function or the joint cumulative distribution of X and Y.

A bivariate function with values $f(x, y)$ defined over xy plane is called **joint probability density function** of two continuous random variables X and Y if and only if

$$P[X, Y \in A] = \int_A f(x, y) dx dy; \text{ for any region in the x y plane.}$$

Similarly, a bivariate function can serve as joint probability density function of a pair of continuous random variable if its values $f(x, y)$ satisfy the conditions

$$i) \quad f(x, y) \geq 0; -\infty < x < \infty; -\infty < y < \infty$$

$$ii) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

Example: Given the joint probability density function of two random variable X and Y

$$f(x, y) = \begin{cases} x(x+y)\frac{3}{5}; & \text{for } 0 < x < 1/2; 1 < y < 2 \\ 0; & \text{elsewhere} \end{cases}$$

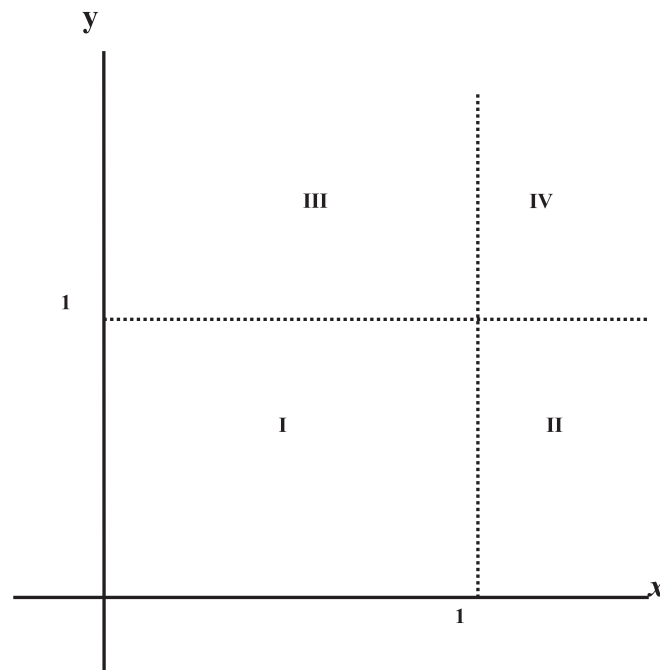


Fig. 27.1: Joint Probability Density Function of X and Y